



UNIT-II: Cloud Service Models

TYBCA • Exam-Oriented PPT • IaaS, PaaS, SaaS & FaaS

Learning Objectives

Explain the meaning and purpose of cloud service/delivery models.

Understand IaaS, PaaS and SaaS through provider–consumer responsibilities.

Compare models based on control, responsibility and purpose.

Explain FaaS as a syllabus supplementary topic.

Prepare structured answers for 5-mark and 8-mark questions.

1. Introduction to Cloud Service Models

A cloud service/delivery model describes what the provider makes available to the consumer.

It indicates how much control and responsibility remain with the consumer.

Cloud Provider → provides cloud services/resources.

Cloud Consumer → uses the services/resources.

Core models: IaaS, PaaS and SaaS.

Primary Cloud Service Models

IaaS — Infrastructure as a Service

PaaS — Platform as a Service

SaaS — Software as a Service

Memory aid: IaaS = Infrastructure; PaaS = Platform; SaaS = Software.

Relationship Among Service Models

SaaS → software/application layer

PaaS → platform/application-delivery layer

IaaS → infrastructure layer

Provider responsibility generally increases from IaaS → PaaS → SaaS.

Consumer control over underlying implementation generally decreases.

2. Infrastructure as a Service (IaaS)

IaaS provides access to infrastructure-based IT resources.

Resources are commonly provided as virtualized infrastructure.

The consumer receives a high level of administrative control.

IaaS forms the infrastructure foundation for higher-level cloud services.

IaaS Resources

Processing — computing capability

Storage — data storage resources

Networking — network connectivity/resources

Hosting — infrastructure/hosting resources

Key idea: IaaS gives the consumer infrastructure to configure and use.

IaaS Responsibilities

Provider: provisions and manages physical infrastructure.

Provider: manages physical processing, storage, networking and hosting.

Consumer: configures virtual infrastructure.

Consumer: installs/manages software and applications.

Consumer has greater administrative responsibility than in PaaS or SaaS.

IaaS — Exam Points

Definition: infrastructure-based IT resources delivered as a cloud service.

High level of consumer administrative control.

Virtualized infrastructure is a key concept.

Resources include processing, storage, networking and hosting.

8-mark structure: Definition → resources → responsibilities → control → diagram/example.

3. Platform as a Service (PaaS)

PaaS provides a pre-defined, ready-to-use environment.

The environment contains pre-packaged products and tools.

It supports application development and application delivery.

The provider manages the platform and underlying infrastructure.

The consumer focuses more on applications than infrastructure.

PaaS Environment

Pre-packaged products and tools

Application development

Application testing

Application deployment

Application/service management

PaaS Responsibilities & Control

Provider: manages the platform and underlying infrastructure.

Consumer: develops, tests and deploys applications.

Consumer has reduced infrastructure-management responsibility.

Consumer has less control over underlying infrastructure than with IaaS.

Key idea: PaaS lets developers focus on application delivery.

PaaS and IaaS

PaaS can be built upon underlying infrastructure.

IaaS provides the infrastructure foundation.

PaaS provides the ready-to-use platform/environment.

Think: IaaS → infrastructure; PaaS → application development environment.

4. Software as a Service (SaaS)

SaaS provides software as a shared cloud service.

The software is made available as a product or utility.

The provider implements and manages the software service.

The consumer primarily uses/configures the service.

Consumer control over implementation is very limited.

SaaS Responsibilities & Control

Provider: implements, manages, maintains and monitors the service.

Consumer: uses the service.

Consumer may configure the service where permitted.

Very limited administrative control over implementation.

Key idea: the consumer uses ready-to-use software rather than managing infrastructure.

SaaS, PaaS and IaaS Relationship

SaaS → application/software layer.

PaaS → platform/environment supporting application delivery.

IaaS → underlying infrastructure.

Layered view: SaaS → PaaS → IaaS → Physical Resources.

5. Function as a Service (FaaS)

FaaS = Function as a Service.

Included here as a syllabus supplementary topic.

It focuses on execution of individual functions.

The developer supplies function code.

The cloud-managed environment executes the function when triggered.

How FaaS Works

1. Developer writes function code.
2. Function is deployed to the cloud function service.
3. An event/request occurs.
4. The function is triggered and executed.
5. The function produces a result.

FaaS: Event-Driven Execution

Functions can be triggered by events or requests.

Execution occurs on demand.

The cloud provider manages the execution environment.

This reduces the need for the developer to manage servers.

FaaS is commonly associated with serverless computing.

FaaS Example: Image Processing

User uploads an image.

File-upload event occurs.

Cloud function is triggered.

Function resizes/processes the image.

Processed image is stored or returned.

6. IaaS vs PaaS vs SaaS

IaaS: infrastructure-based resources; higher consumer control.

PaaS: ready-to-use platform/environment; focus on application development.

SaaS: shared software service; focus on using/configuring software.

Provider responsibility increases from IaaS → PaaS → SaaS.

Consumer responsibility and underlying control generally decrease.

IaaS vs PaaS

IaaS = infrastructure; PaaS = platform.

IaaS gives more infrastructure control.

PaaS reduces infrastructure management.

IaaS: consumer manages more of the environment.

PaaS: consumer focuses on application development and deployment.

PaaS vs SaaS

PaaS = build/deploy applications.

SaaS = use a ready-made software service.

PaaS gives more application-level control.

SaaS provides a finished service with limited implementation control.

Key difference: application development vs application usage.

SaaS vs FaaS

SaaS provides a complete software service.

FaaS executes an individual function.

SaaS is primarily used through an application/service.

FaaS is commonly triggered by an event/request.

Key difference: complete software service vs function execution.

Overall Service Model View

IaaS → Infrastructure

PaaS → Platform

SaaS → Software

FaaS → Function execution (supplementary topic)

Provider responsibility generally increases from IaaS → PaaS → SaaS.

Exam-Oriented Answer Pattern

8 marks: Definition → Explanation → Components/Characteristics.

Then write provider responsibilities and consumer responsibilities.

Mention level of consumer control.

Add a simple labelled diagram and suitable example.

For comparison questions: use a clear table followed by a short conclusion.

Important 8-Mark Questions

1. Explain IaaS with its resources, responsibilities and level of consumer control.
2. Explain PaaS with its environment, responsibilities and level of consumer control.
3. Explain SaaS with its responsibilities and level of consumer control.
4. Compare IaaS, PaaS and SaaS.
5. Explain FaaS, its working and event-driven execution with an example.
6. Explain the relationship among IaaS, PaaS and SaaS.

Quick Revision

IaaS → Infrastructure → High consumer control

PaaS → Platform → Application development/deployment

SaaS → Software → Ready-to-use service

FaaS → Function → Event-driven execution

Remember: provider responsibility generally increases from IaaS → PaaS → SaaS.

Unit-II: Key Takeaway

Cloud service models define what the provider delivers and what the consumer manages.

IaaS provides infrastructure.

PaaS provides a development/delivery environment.

SaaS provides ready-to-use software.

FaaS provides event-driven function execution as a supplementary syllabus topic.